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Real Madrid CF

Tactical & Player Analytics Dashboard

Transforming event-level football data into actionable tactical intelligence

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Escuela Universitaria Real Madrid — Universidad Europea

Python · Plotly/Dash · Unicorn · Render · Data: Opta Stats Perform (event-level)

The Problem & The Objective



- Elite football generates ~1,500–2,000 on-ball events per match — far too granular to read by eye.
- Traditional analysis suffers from outcome bias: a scoreline hides whether a performance was earned.
- Coaching staff need the data-to-decision layer that turns a raw Opta feed into tactical answers.
- Objective: quantify HOW Real Madrid play — not just whether they win — across attacking, defending, transitions and set pieces.
- Deliver an interactive, transparent, reproducible tool for match prep, opposition scouting and review.

KEY POINT

Every metric is traceable to an explicit formula over observed events — no synthetic data anywhere.

Dashboard Architecture — Seven Modules



- Overview — season KPIs, results table, xG / xGA trends, goals trend, form tracker.
- Match Analysis — shot maps, xG chart, pass map, pass network, shot zones, Zone 14, GK distribution, crossing patterns, goalmouth, defensive actions, transitions, set pieces.
- Player Analysis — squad table, per-player stats, multi-radar, touch heatmaps.
- Tactical Phases — PPDA, pressing maps, four-phase A/B/C/D scoring, recoveries, field tilt.
- Opponent Scout — opposition profiling, threat heatmap, set-piece tendencies, auto report.
- Benchmarking — cross-competition and cross-season comparisons.
- Report — one-click PDF / DOCX export from any selection.

KEY POINT

All seven modules share one data pipeline and one set of KPI definitions.

- Source: Opta Stats Perform event-level JSON — one file per match.
- Scope: 108 clean match files · 3 competitions (LaLiga · UCL · Copa del Rey) · 2 seasons.
- Pipeline: parse & normalise events → compute match KPIs → re-slice to player, phase, opponent.
- Transparent models: positional xG (location), PPDA (pressing), field tilt, four-phase A/B/C/D.
- Showcase fixture for the visuals that follow: Real Madrid 2–1 Barcelona, LaLiga 2025–26 MD10.

KEY POINT

Possession is a documented pass-share proxy; no tracking data — limitations stated openly.

Multi-Match Analysis Mode

**NEW****Multi-Match (up to 5)**

Select any 5 specific matches for aggregated cross-match analysis.

HOW**Mode Selector**

Dropdown now has 3 options: Single Match · Match Range · Multi-Match (≤ 5).

DATA**Shared Store Relay**

dcc.Store propagates selected paths to all 25+ callbacks without performance cost.

SCOPE**All Sections Covered**

Shot map, xG, pass network, Zone 14, transitions, set pieces — all multi-match aware.

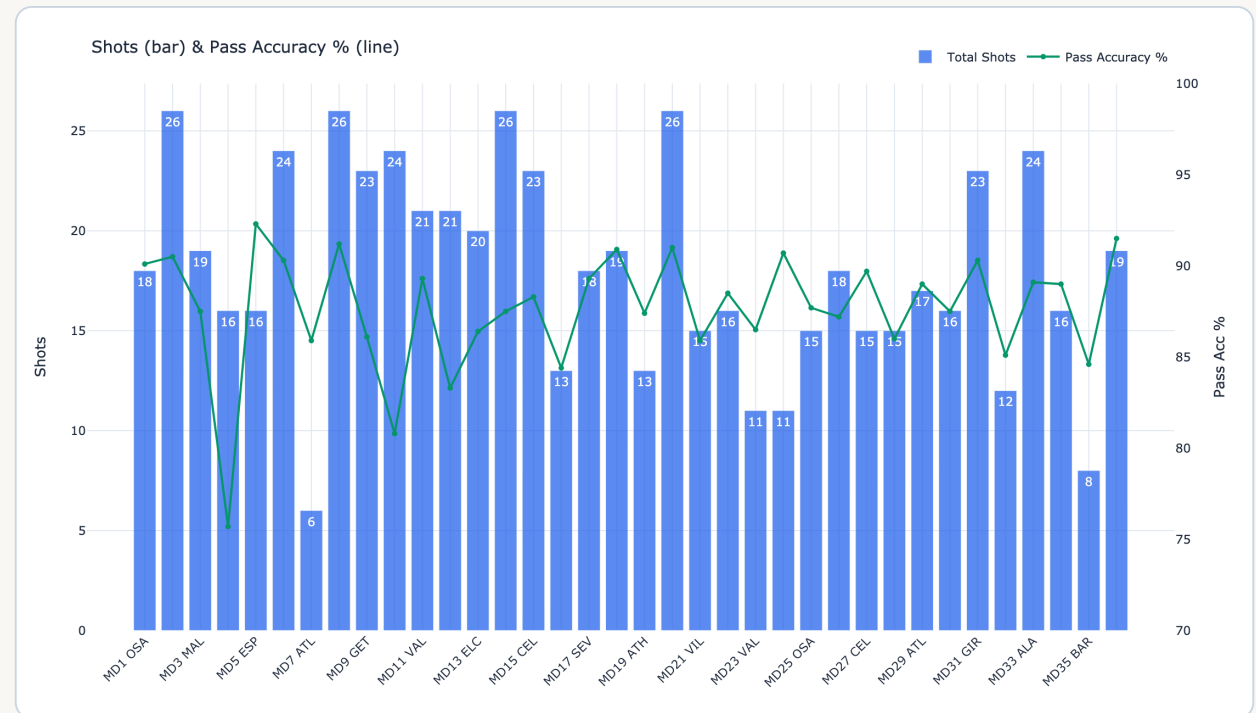
WHY IT MATTERS

- Range mode aggregates a continuous window of matches.
- Multi-Match lets an analyst cherry-pick: home legs, rivals, or run of form.
- Same visualisations, aggregated data — no new learning curve.
- Enables direct pattern comparison across non-consecutive fixtures.
- Implemented without breaking Single or Range modes.

Overview — Season Context



- Tracks process metrics across the season, not just results.
- Process metrics are more stable and more predictive than outcomes.
- Falling shots with high pass accuracy = early warning of sterile possession.
- The panel a coach watches weekly to spot a problem before the table shows it.

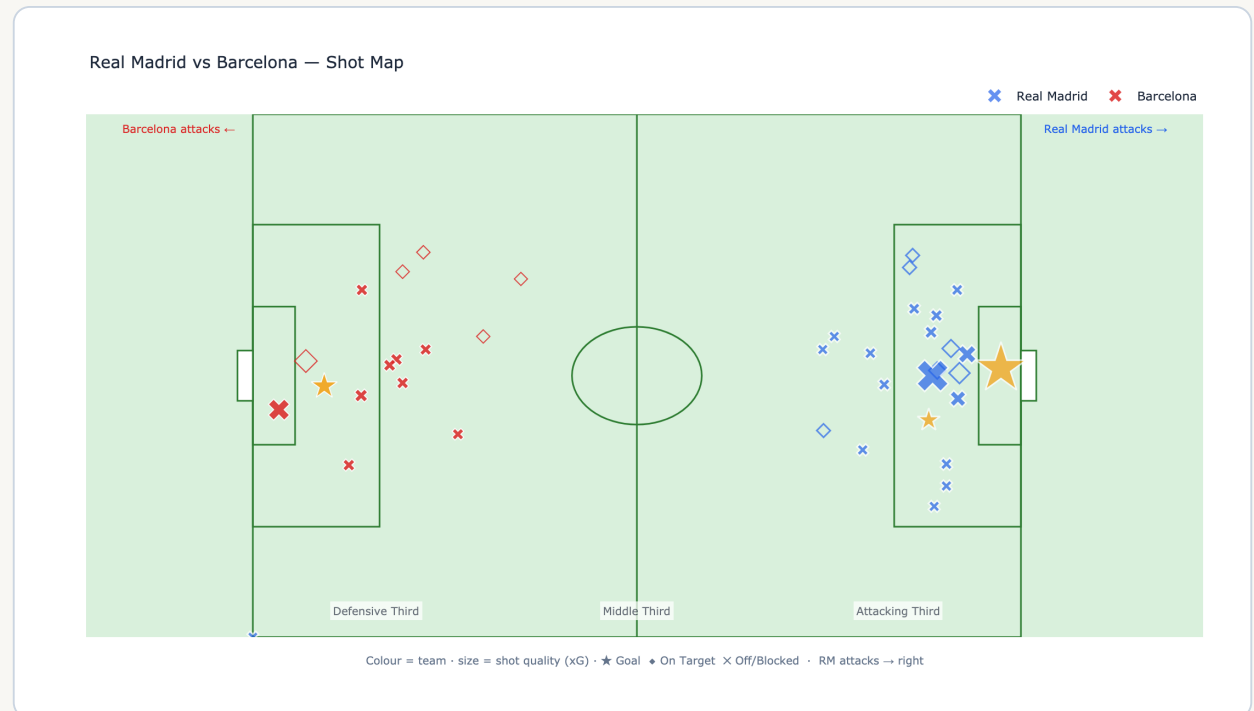


Overview module — Shots & Pass-Accuracy Trend, live from the real LaLiga 2025–26 feed.

Match Analysis — Shot Map



- Every shot at its true location; marker size $\propto$ xG;
★ = goal.
- Real Madrid (blue) attack right; Barcelona (red) attack left.
- RM's chances cluster centrally inside the box — high volume AND high quality.
- Answers: where, and how good, were the chances — for both teams.

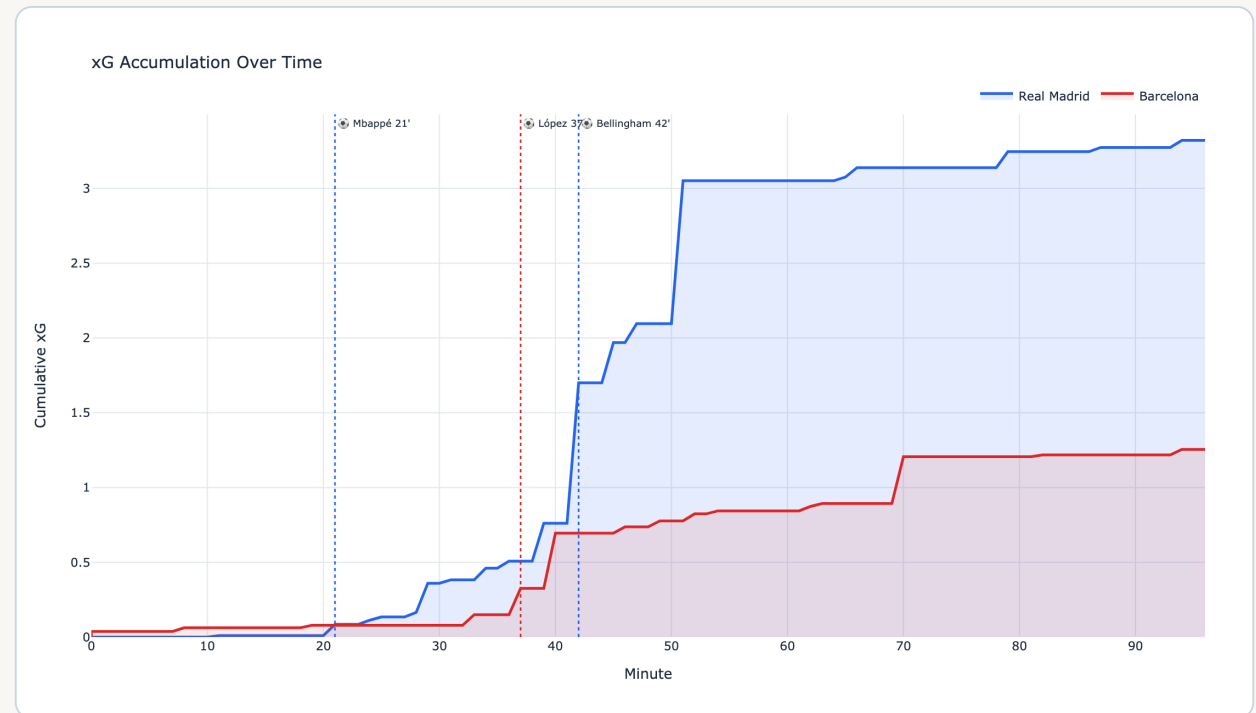


Match Analysis — Shot Map · Real Madrid 2-1 Barcelona (24 shots · 2 goals · 3.34 xG).

Match Analysis — xG Accumulation



- Running total of Expected Goals, minute by minute, for both teams.
- Slope = rate of chance creation; a tall step = a single large chance.
- Reveals WHEN control happened — early dominance vs a nervy finish.
- Full-time gap between the lines = the match's xG difference.

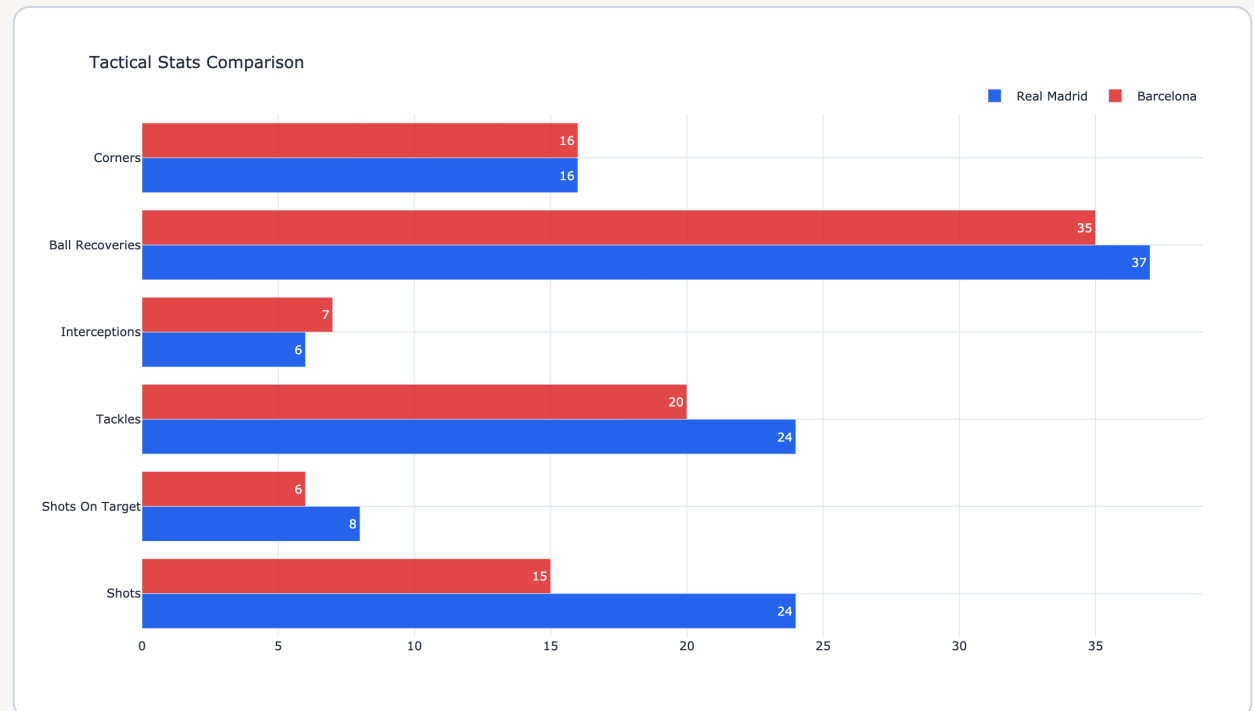


Match Analysis — Expected Goals accumulating through the match from the real event feed.

Match Analysis — Tactical Comparison



- Head-to-head bars of headline KPIs: possession, shots, passing, defensive actions.
- Built for communication speed — 'who won which battle' in one glance.
- Each metric read in relative context — the gap to the opponent, not just the raw value.

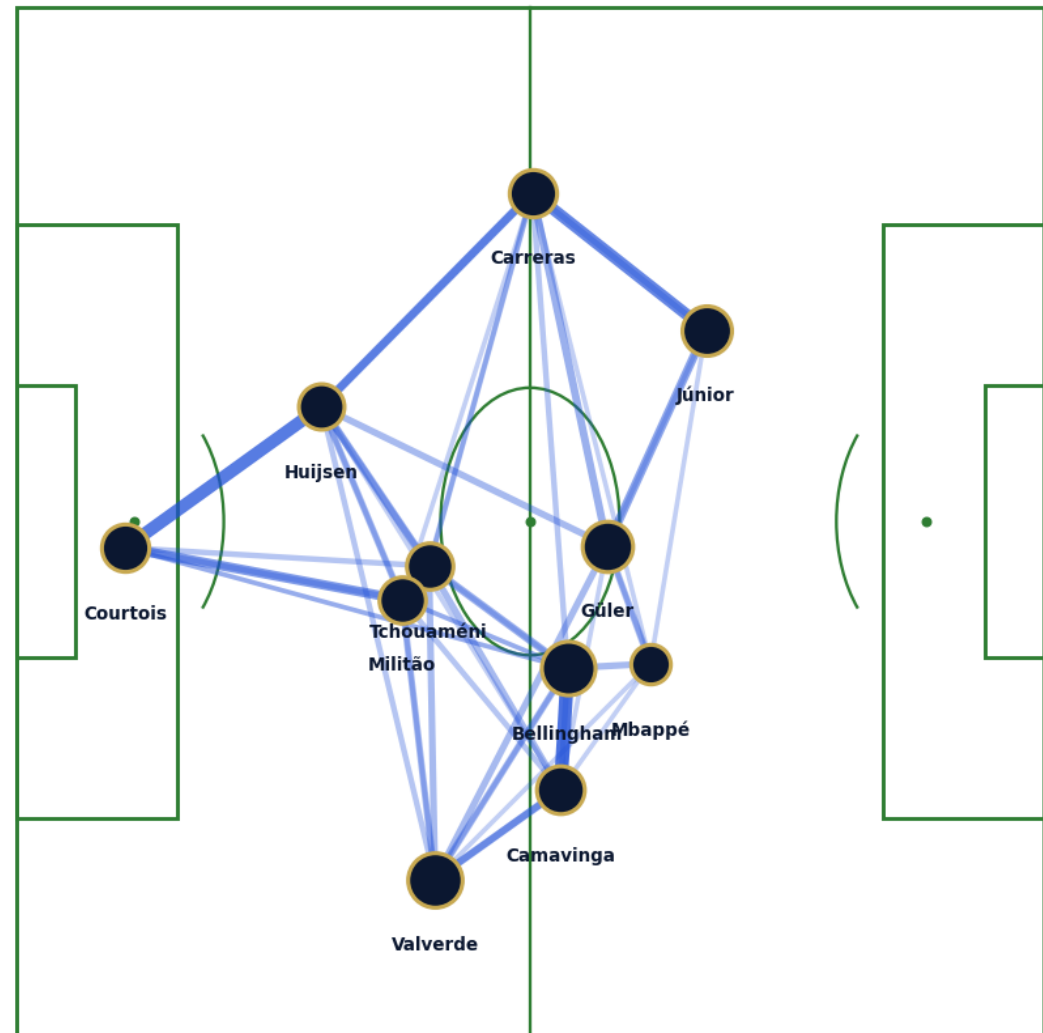


Match Analysis — Tactical Comparison, Real Madrid vs Barcelona, from the real feed.

Match Analysis — Pass Network

- Directed graph of the starting XI's passing connections.
- Node position = player's average event location on the pitch.
- Node size = total touches; line width & opacity = pass frequency.
- Exposes the dominant playmaking triangle and width of ball circulation.
- High centrality in Zone 14 = chance-creating hub detected automatically.

Pass Network — Real Madrid Starting XI
Node size = touches · line width & opacity = pass frequency

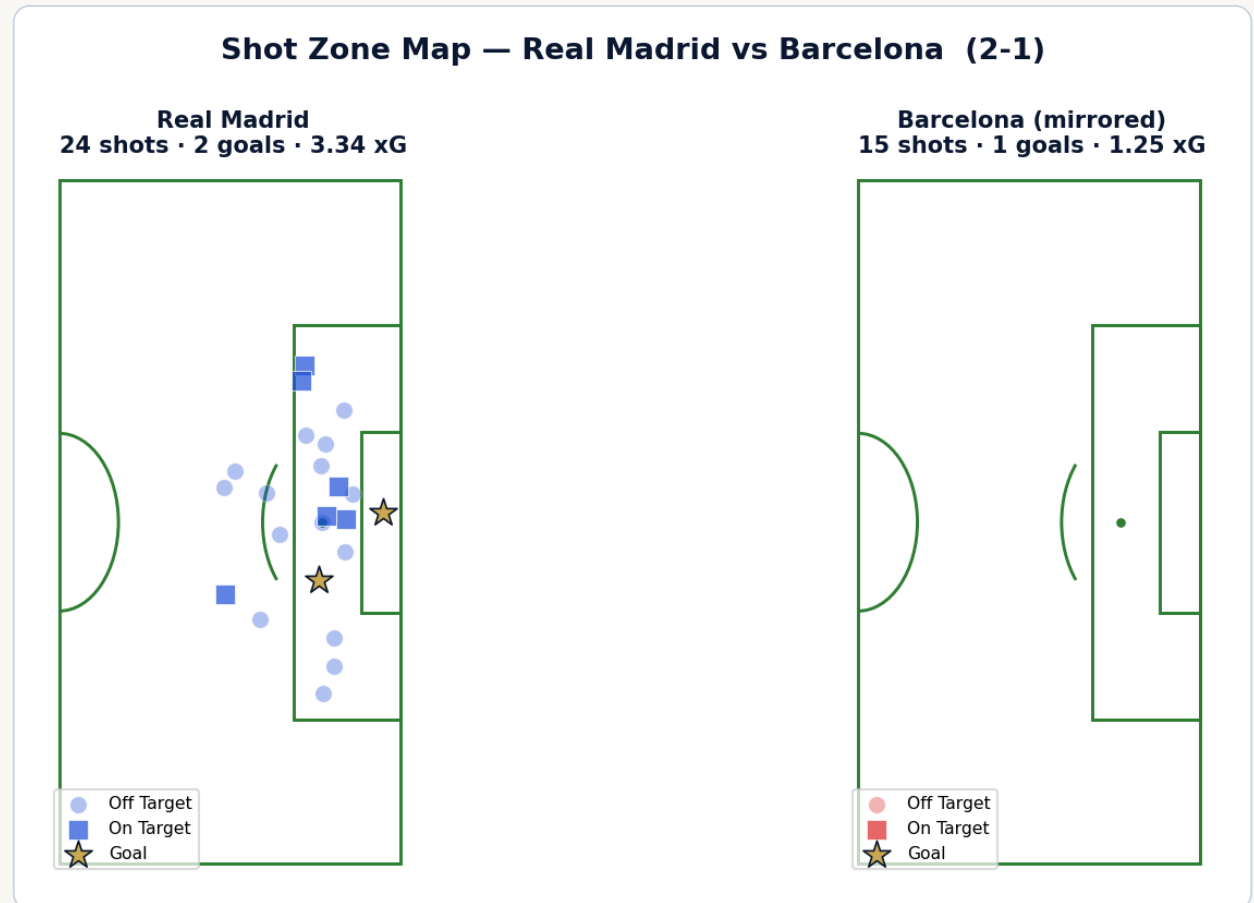


Pass Network — inferred from Opta event sequences for the Real Madrid starting XI.

Match Analysis — Shot Zone Map



- Every shot plotted at its true Opta coordinate on a half-pitch view.
- Outcome coded by shape: ★ = goal · ■ = on target · ● = off target / blocked.
- Side-by-side: Real Madrid (left) and opponent (right, mirrored to same direction).
- Reveals WHERE both teams created chances — box dominance vs long-range reliance.



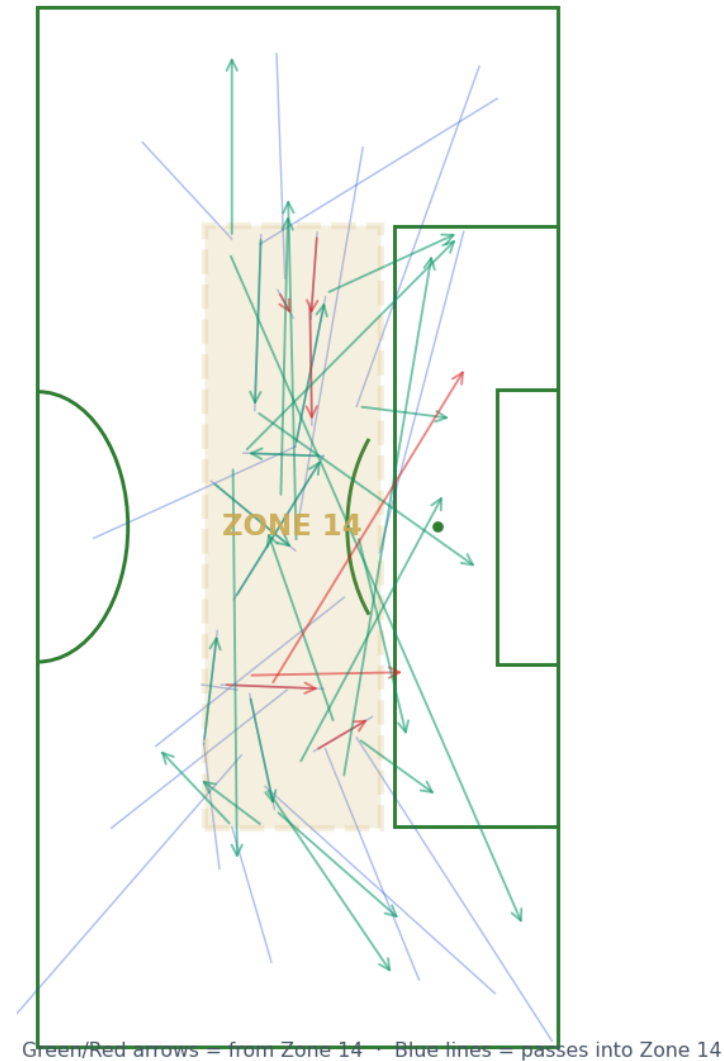
Shot Zone Map — spatial shot quality comparison for both teams, from the real event feed.

Match Analysis — Zone 14 Passing



- Zone 14 (the half-space between penalty area top and wide channels) is where chances originate.
- Green arrows = completed passes from Zone 14; red = unsuccessful.
- Blue lines = passes played INTO Zone 14 (feeding the danger zone).
- High volume into Zone 14 + high completion from it = dangerous central build-up.

Zone 14 Passing — Real Madrid 32 passes from Zone 14 (25 completed) · 41 passes into Zone 14

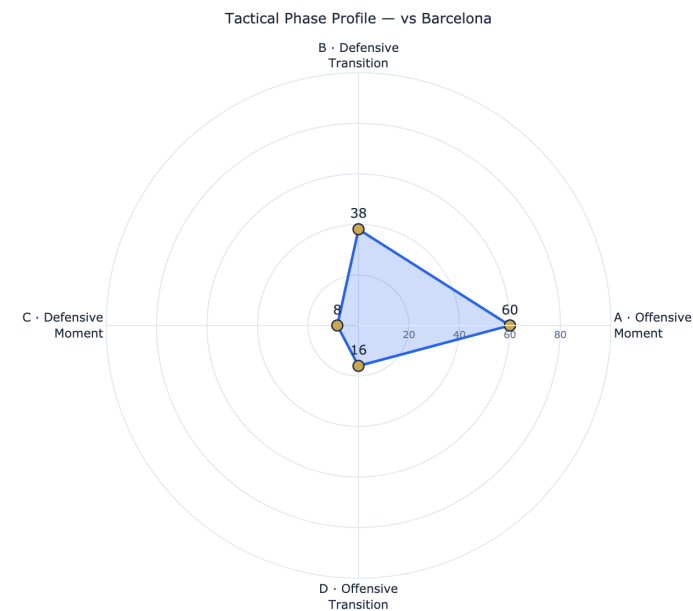


Zone 14 Passing — Opta passes filtered to the central attacking half-space.

Tactical Phase Profile (A/B/C/D)



- Four moments of play scored 0-100: Offensive Moment (A), Defensive Transition (B), Defensive Moment (C), Offensive Transition (D).
- The radar 'shape' is the match's tactical fingerprint.
- This fixture: strong A + B, lower C + D = possession-and-counter-press.
- Composite indices grounded in PPDA, xG, recoveries & transition rate.

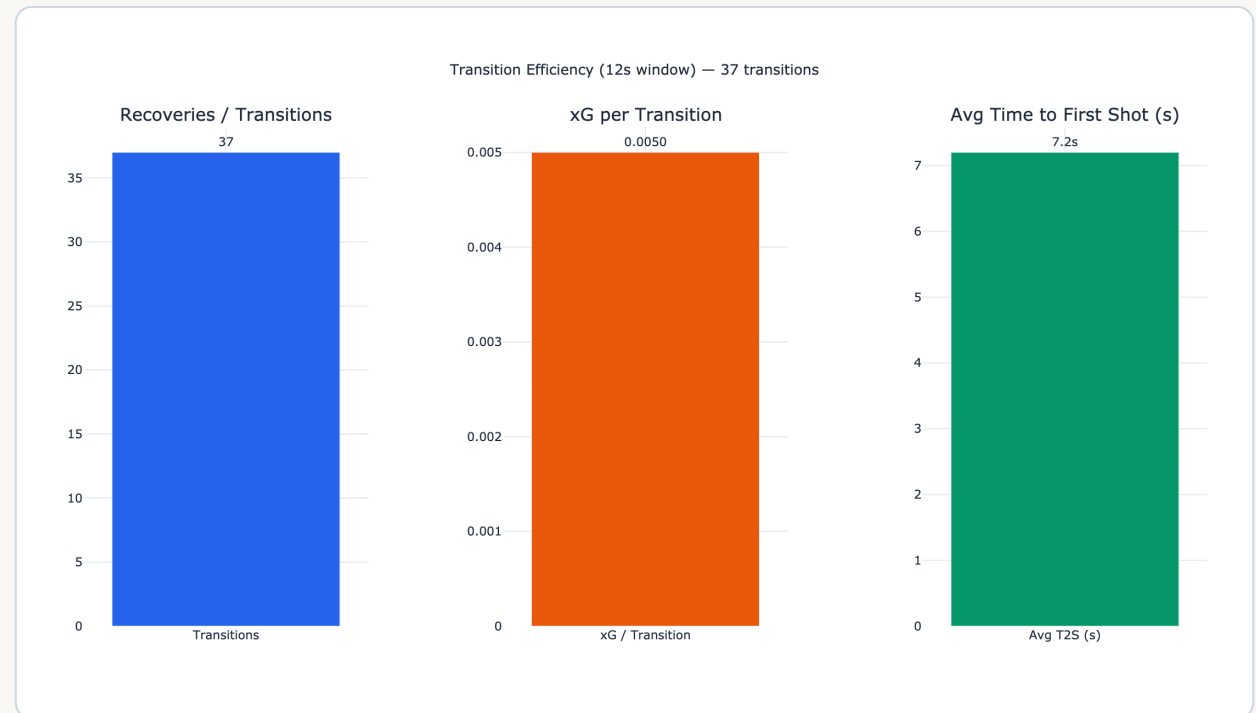


Tactical Phase Profile — the dashboard's executive summary of how the team played.

Attacking Transitions



- Locates every ball regain, then tests for a shot within the time window (5-15 s).
- Quantifies fast-break efficiency and transition xG — the counter-attack as a number.
- Against possession sides, transitions punish their committed players.
- Window is an analyst control, not a hidden assumption.

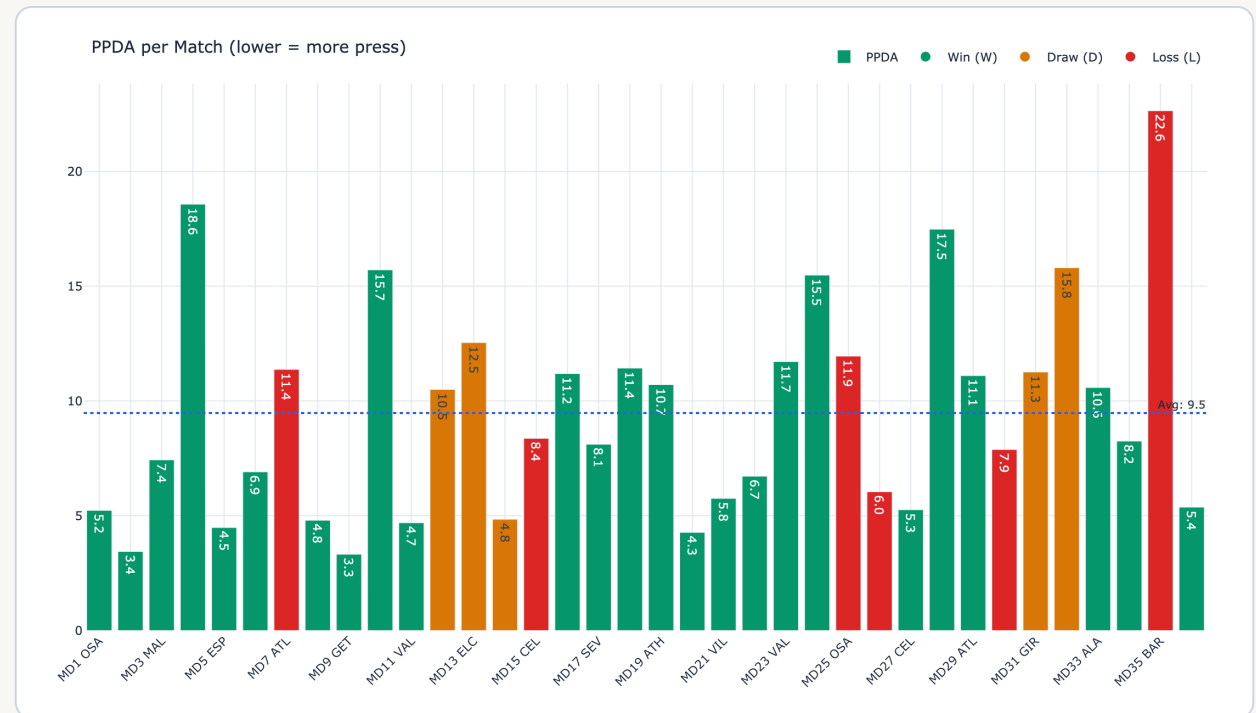


Match Analysis — Transition Metrics, quantifying Real Madrid's counter-attacking threat.

Pressing Intensity — PPDA Trend



- PPDA = passes the opponent completes per RM defensive action (lower = more intense).
- Plotted every matchday with the season average marked.
- Tight band = a drilled pressing identity; wide scatter = game-state dependent.
- The instant verdict on whether 'we pressed well today' is actually true.

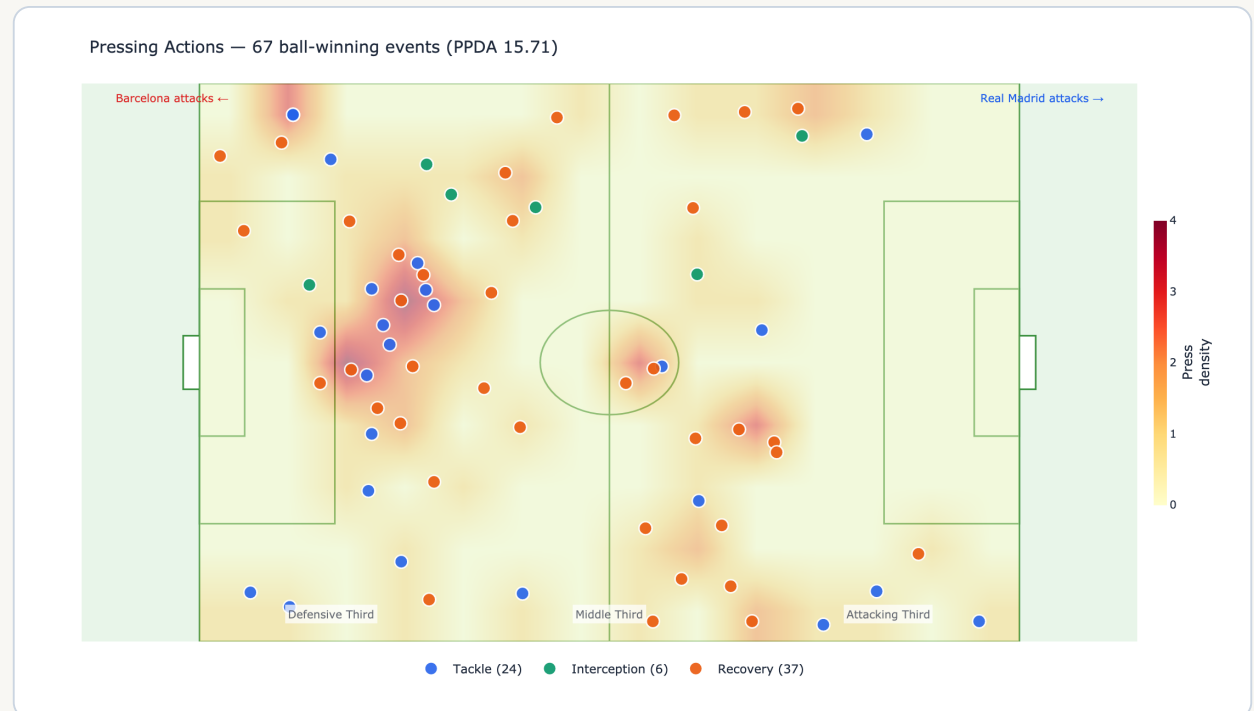


Tactical Phases — season-long PPDA trend across the real LaLiga campaign.

Pressing Actions Map



- Where on the pitch Real Madrid engaged defensively — every press, tackle, interception.
- Reveals pressing HEIGHT: high press vs mid-block vs low block.
- Exposes asymmetry — a flank pressed harder is a flank the opponent will exploit.
- Direct evidence of whether the pressing instruction was executed.

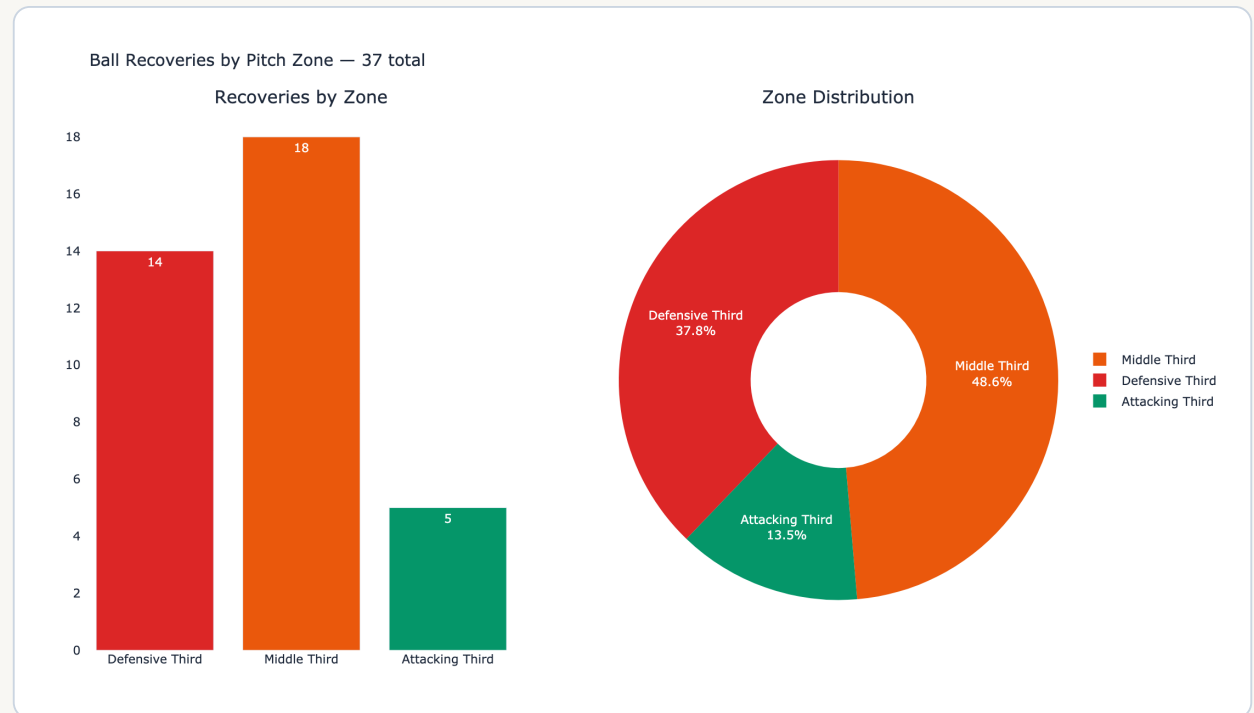


Tactical Phases — Pressing Actions Map for the real match.

Ball Recoveries by Zone



- WHERE the team wins the ball back — defensive / middle / attacking third.
- This match: 37 recoveries — Mid 48.6% · Def 37.8% · Att 13.5% (mid-block profile).
- Attacking-third recoveries are the most dangerous regains in football.
- Read against intention: great for a mid-block, disappointing for a high press.

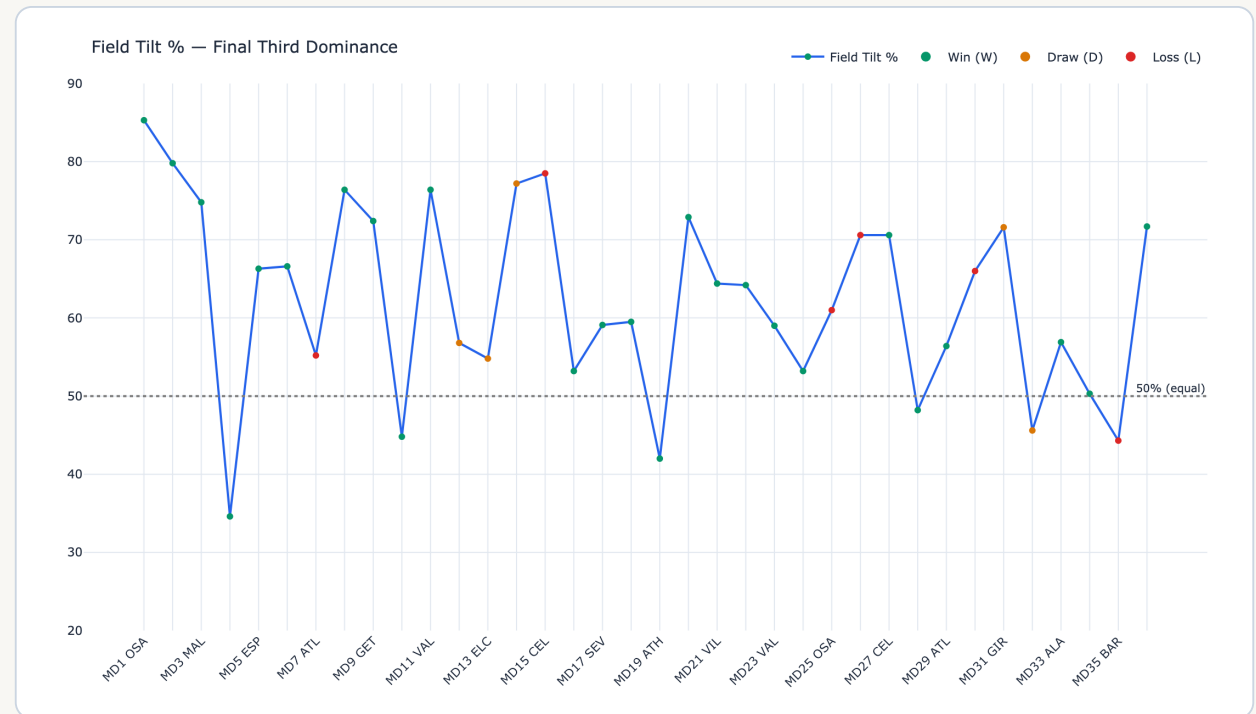


Tactical Phases — Ball Recoveries by Pitch Zone from the real feed.

Territorial Dominance — Field Tilt



- Share of all final-third activity belonging to Real Madrid.
- Antidote to misreading possession — control that becomes territory.
- 55% possession with 75% field tilt beats 60% possession with 51% tilt.
- Validates the PURPOSE of keeping the ball.

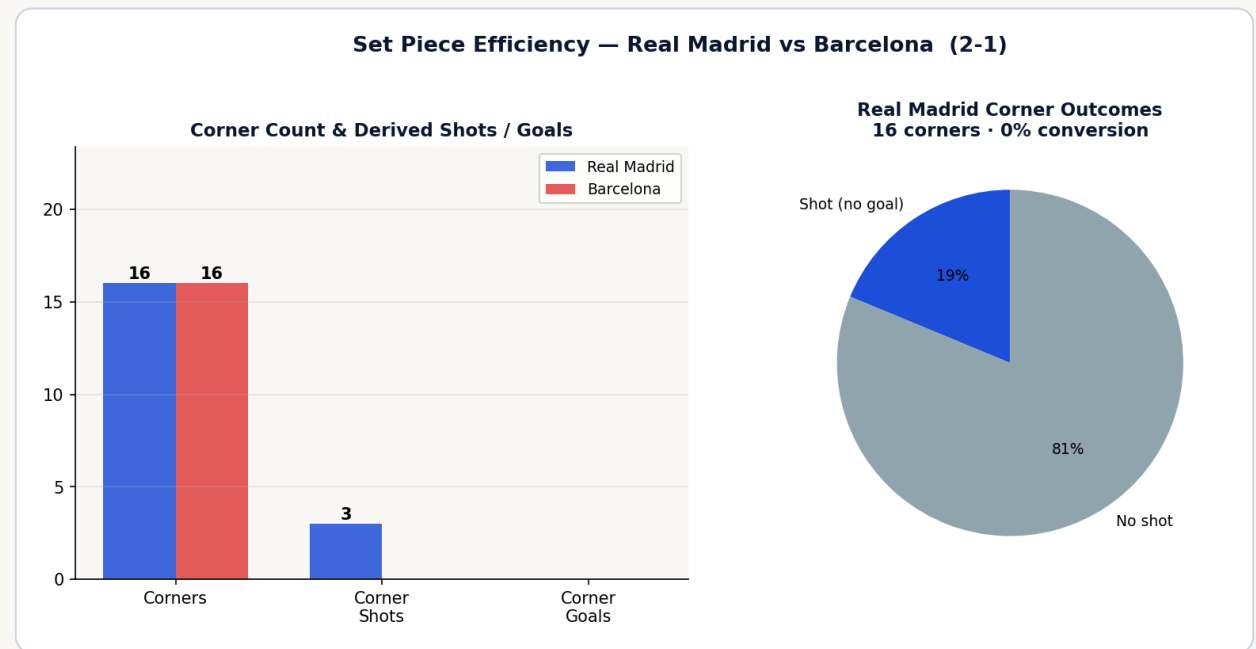


Tactical Phases — Field Tilt across the real season.

Set Piece Efficiency



- Counts corners awarded, shots derived from corners and goals from corners.
- Pie chart shows RM's corner conversion funnel: corners → shots → goals.
- Side-by-side bars compare RM and opponent set-piece output.
- Also covers free kicks and penalty areas in the full dashboard.
- Set pieces account for ~30% of goals at elite level — this panel quantifies them.



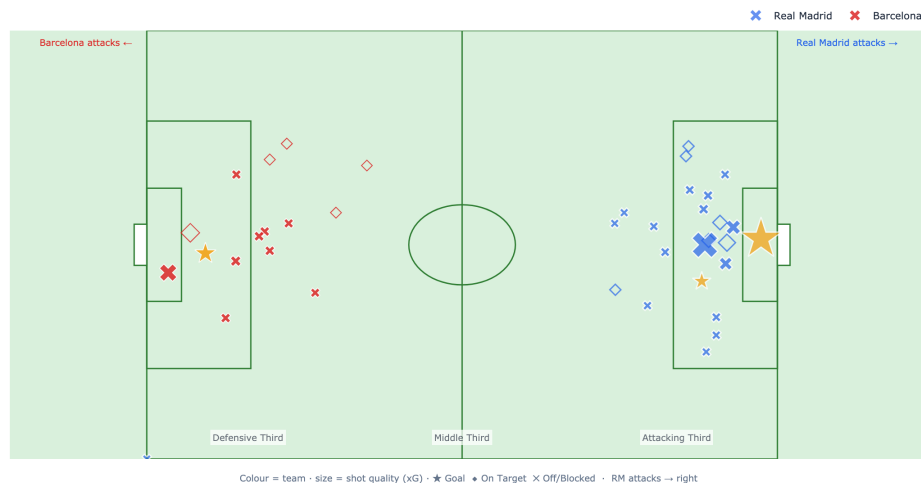
Set Piece Efficiency — derived from Opta event sequences for the real match.

How the Panels Triangulate



- No single panel proves a conclusion — together they make it unarguable: the Shot Map shows high-quality central chances; the radar collapses the match into one possession-and-counter-press shape.
- Triangulation runs across xG, field tilt, recoveries and PPDA — a tactical claim that is true leaves its fingerprint on several independent panels at once.

Real Madrid vs Barcelona — Shot Map



Tactical Phase Profile — vs Barcelona



Two independent views of the same real match converging on one tactical read.

OUTPUT**PDF & DOCX Export**

One-click branded report in PDF or DOCX — downloads directly from the browser.

SCOPE**Match Scope**

Mirrors Match Analysis: Single, Range or Multi-Match (up to 5).

PLAYER**Player Breakdown**

Per-player stats table auto-populates from squad in selected matches.

CONTENT**9 Sections**

Filter Summary · Executive Summary · Team · Player · Tactical · Match · Benchmarking · Appendix · Conclusion.

REPORT SECTIONS

- 01 — Filter Summary (competition, season, matches).
- 02 — Executive Summary (W/D/L, xG, PPDA, possession).
- 03 — Team Performance (aggregate + match-by-match table).
- 04 — Player Performance (goals, shots, passes, xG per player).
- 05 — Tactical Analysis (pressing maps, position maps).
- 06 — Match Analysis (shot maps, xG chart, pass network).
- 07 — Benchmarking (league comparisons).
- 08 — Visualisation Appendix (all captured charts).
- 09 — Conclusion: Pre / Match / Post-match evaluation.

- All nine core analysis features IMPLEMENTED with dedicated panels:
 - Pre-Match · Attacking Transitions · Organised Attack · Defensive Strategy · Set Pieces.
 - Possession Recovery · Post-Match · Final-Third Entries · Match Info.
- New in June 2026: Multi-Match mode · Pass Network · Shot Zone Map · Zone 14 · Set Piece Efficiency · GK Distribution · Crossing Patterns · Goalmouth Map · Defensive Actions Map · Report Generator.
- Deployed publicly via GitHub → Render CI/CD (Plotly/Dash + Unicorn).
- Report export: PDF and DOCX with UE / Real Madrid branding.

KEY POINT

108 clean match files across 3 competitions and 2 seasons; no synthetic data anywhere.

Limitations & Future Scope



- Event data only — no tracking: off-ball shape & pressing distances are approximated.
- xG is a transparent positional model (location only), not a provider post-shot model.
- Phase scores are interpretive 0–100 composites, not physical units.
- Descriptive & diagnostic — not predictive, not real-time.
- Roadmap: tracking-data integration · player per-90 depth · Expected Threat (xT) · ML pattern clustering · predictive analytics · video integration · live dashboards.

KEY POINT

The platform is a foundation for continued research and professional deployment.



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From Data to Tactical Decision

One transparent, interactive platform — match prep, scouting, review and player development.

Real Madrid CF Tactical Dashboard

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